

ITE2953 - Programming Group Project 25S2

Project Proposal

Local Tourist Day-Visit Planner and Information System

(Kandy, Sri Lanka)

Submitted by:

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Abbreviations

Abbreviation	Meaning
API	Application Programming Interface
BA	Business Analysis / Business Analyst
CSS	Cascading Style Sheets
EF Core	Entity Framework Core
FR	Functional Requirement
HTML	Hypertext Markup Language
IEEE	Institute of Electrical and Electronics Engineers
JSON	JavaScript Object Notation
NFR	Non-Functional Requirement
ORM	Object-Relational Mapping
REST	Representational State Transfer
SDLC	Software Development Life Cycle
SPA	Single Page Application
SQL	Structured Query Language
UI	User Interface
UNESCO	United Nations Educational, Scientific and Cultural Organization

1.0 Introduction / Background

Kandy is one of Sri Lanka's most popular tourist destinations, well known for its rich cultural heritage, religious significance, and scenic landscapes. Notable landmarks within a 25 km radius include the Sri Dalada Maligawa (Temple of the Sacred Tooth Relic) [1], the Royal Botanical Gardens, Udawattakele Forest Sanctuary [2], and the Ceylon Tea Museum, which together attract large numbers of local and international visitors every year.

Despite this popularity, tourists often struggle to find accurate, well-organized information about attractions in the area. Details are scattered across multiple websites and social media pages, making it time-consuming to compare destinations, locate nearby places, and plan a one-day itinerary. Many visitors also lack easy access to essential details such as opening hours, travel tips, and distances. From a business perspective, this represents a clear and recurring information gap in the local tourism sector. To address this problem, this project proposes a web-based Local Tourist Day-Visit Planner and Information System for Kandy that centralizes tourist attraction information and allows users to build a personalized one-day travel plan.

1.1 Problem Statement

Tourists visiting Kandy often find it difficult to obtain accurate, reliable, and well-organized information about local attractions within a 25 km radius, since details such as opening hours, travel tips, and distances are scattered across multiple websites and social media platforms. This makes it challenging to compare destinations and plan an efficient one-day visit. From a business analysis standpoint, the absence of a single, trusted source of tourist information is an operational inefficiency that directly affects the visitor experience; a centralized web-based system is therefore needed to give tourists comprehensive, up-to-date information and the ability to conveniently organize their day trips.

1.2 Aim

To design and develop a web-based Local Tourist Day-Visit Planner and Information System that enables tourists to easily explore, access, and organize information about tourist attractions within a 25 km radius of Kandy for efficient one-day trip planning.

1.3 Project Objectives

The main objective of this project is to build a centralized, user-friendly platform that helps tourists plan a one-day visit to Kandy. The specific objectives are:

1. To develop a web-based application that provides comprehensive information about tourist attractions within a 25 km radius of Kandy.
2. To enable tourists to search and filter tourist attractions based on categories such as religious sites, heritage locations, nature, museums, viewpoints, and recreational places.
3. To provide detailed information for each tourist attraction, including descriptions, opening hours, travel tips, distances, and map locations to support informed travel decisions.
4. To allow users to create a personalized one-day visit plan by selecting multiple tourist attractions based on their interests and available time.
5. To develop an administrator module that allows administrators to add, update, and delete tourist attraction information, ensuring that the system remains accurate and up to date.
6. To create a user-friendly and responsive web application that improves tourists' access to reliable local tourism information and enhances their overall travel planning experience.

1.4 Scope and Limitations

This section clearly defines what the proposed system will include (scope) and will not include (limitations).

1.4.1 Scope

- Displaying a minimum of 15 tourist attractions located within approximately 25 km of Kandy.
- Categorizing tourist attractions into groups such as Religious & Heritage, Nature, Adventure, Museum, Viewpoint, Recreation, and Eco Tourism.
- Allowing users to search and filter tourist attractions based on category or place name.
- Displaying detailed information for each attraction, including descriptions, opening hours, travel tips, approximate distance, images, and map location.
- Allowing tourists to create a personalized one-day visit plan by selecting preferred attractions.
- Providing an administrator module to add, edit, and delete tourist attraction information.
- Developing a responsive web application that can be accessed using desktop and mobile web browsers.

1.4.2 Limitations

- The system only includes tourist attractions located within approximately 25 km of Kandy.
- Online hotel booking services are not supported.
- Online transportation or vehicle reservation facilities are not included.
- Online ticket booking and payment processing are outside the project scope.
- User registration and online travel package management are not included.
- Tourist information depends on the data maintained by the system administrator and may require periodic updates.

1.5 Target Users

The proposed system is designed to serve two primary user groups: tourists and system administrators. Each group has different responsibilities and interacts with the system based on its specific needs.

Tourist Users: Tourist users are the primary users of the system. They may be residents, domestic travelers, or international visitors who wish to explore tourist attractions within a 25 km radius of Kandy. The system enables them to search and filter attractions by category, view detailed information about each destination, access location maps, and create a personalized one-day travel itinerary.

System Administrator: The system administrator is responsible for managing and maintaining the information available in the application. The administrator can add new tourist attractions, update existing information, and remove outdated or incorrect records, ensuring that the system always provides accurate and up-to-date information.

1.6 Stakeholder Identification Plan

As part of the requirement elicitation process, the proposed system was discussed with four group members acting as representative peer stakeholders, through a structured online consultation forum, to understand the information and features that tourists require when planning a one-day visit to Kandy. Each stakeholder led one discussion thread within the forum. Their feedback, summarized in Table 1 below, was analyzed and used to identify the functional requirements and key features of the system (see Appendix A for the full consultation record).

Stakeholder	Registration No.	Role in Consultation	Key Requirement Identified
James JBK	E2410439	Peer Stakeholder - Led Discussion 1	Identified that tourist information is scattered across multiple sources and requested that each attraction list include essential details such as name, description, photos, opening hours, entrance fees, and contact information.
Seema AFZ	E2410442	Peer Stakeholder - Led Discussion 3	Recommended integrating Google Maps to display each attraction's exact location, distance, and estimated travel time, and confirmed the need for search, category filtering, and one-day itinerary planning.
Werasekara SNDRWMMGJCB	E2410436	Peer Stakeholder - Led Discussion 4	Suggested category-based filtering with practical planning details (visit duration, ratings, best time to visit, nearby transport), and recommended keeping the system tightly scoped to single-day visit planning.

Madhushan S	E2410438	Peer Stakeholder - Led Discussion 5	Suggested displaying nearby facilities (restaurants, parking, washrooms, emergency services) and recommended an administrator panel to keep attraction information accurate and up to date.
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Table 1: Stakeholder Identification Plan

Note: Krishalani Thiyagaraja (E2410433) is the developer of this individual project and is therefore not counted as one of the four consulted stakeholders. Full evidence of the consultation, including Discussions 1-6, is provided in Appendix A.

The feedback collected from the stakeholders was analyzed and incorporated into the proposed system requirements to ensure that the application meets the needs of its intended users.

1.7 Proposed Solution and System Overview

To address the problem identified in Section 1.1, this project proposes the development of a centralized, web-based Local Tourist Day-Visit Planner and Information System for Kandy. From a business perspective, the proposed solution removes the inefficiency of tourists having to consult multiple, inconsistent sources by consolidating verified attraction information, search and filter capabilities, and personalized itinerary planning within a single platform.

The solution follows standard three-tier web architecture. An Angular single-page application (SPA) provides a presentation layer and user interface. An ASP.NET Core Web API backend, implemented in the C# programming language, forms the application layer, applying business logic and exposing RESTful endpoints. A SQL Server database forms the data layer, storing tourist attractions, category, and administrator data. The Angular frontend communicates with the backend over HTTP using JSON, consistent with the standard client-server model in which the browser sends a request that is processed by the server before results are returned for display.

Key Features:

- Display tourist attractions within approximately 25 km of Kandy.
- Search tourist attractions by place name and filter by category.
- View detailed attraction information, including descriptions, opening hours, travel tips, approximate distance, images, and map location.
- Display attraction locations on an interactive map.
- Create a personalized one-day visit plan by selecting preferred attractions.
- Secure administrator login with the ability to add, update, and delete tourist attraction records.
- Responsive interface compatible with desktop and mobile web browsers.

Figure 1 below presents a simple layered block diagram summarizing the proposed system architecture.

Layer	Component	Technology	Responsibility
Presentation	Angular Single Page Application	Angular, TypeScript, HTML5, CSS3	Renders the user interface and handles user interaction (search, filter, itinerary builder, administrator forms).
Application / API	ASP.NET Core Web API	C#, ASP.NET Core, RESTful API	Implements business logic, exposes RESTful endpoints, and enforces authentication and authorization.
Data	SQL Server Database	SQL Server, Entity Framework Core	Stores tourist attraction records, categories, and administrator accounts.
External Service	Google Maps Integration	Google Maps API	Provides interactive maps, distances, and location information for each attraction.

Figure 1: Proposed System Architecture (Layered View)

1.8 Preliminary List of Places

The following table presents the preliminary list of tourist attractions that will be included in the proposed system. These attractions are located within approximately a 25 km radius of Kandy and represent a variety of categories to provide tourists with diverse travel experiences.

No.	Place Name	Category	Approx. Distance from Kandy	Description
1	Royal Botanical Gardens, Peradeniya	Nature	6 km	A world-renowned botanical garden featuring a diverse collection of plants, orchids, and landscape gardens.
2	Hanthana Katusukonda Hike, Tea & Waterfall	Adventure / Nature	8 km	A scenic hiking destination offering tea plantations, waterfalls, and panoramic mountain views.
3	British Garrison Cemetery	Heritage	1 km	A historic cemetery preserving the graves of British colonial officers and civilians.
4	Sir James Taylor's Loolkandura Tea Heritage Hike	Heritage / Nature	20 km	A historic tea estate associated with the pioneer of the Sri Lankan tea industry.
5	Udawattakele Forest Sanctuary	Nature	2 km	A protected forest reserve known for its biodiversity, walking trails, and birdlife.
6	Ceylon Tea Museum	Museum	4 km	A museum that showcases the history and development of

				Sri Lanka's tea industry.
7	Bellwood View Point	Viewpoint	15 km	A scenic viewpoint offering panoramic views of the surrounding mountains and valleys.
8	Water World Kandy	Recreation	9 km	A recreational attraction featuring aquatic exhibits suitable for families and visitors.
9	Sri Dalada Maligawa (Temple of the Sacred Tooth Relic)	Religious & Heritage	0.5 km	Sri Lanka's most sacred Buddhist temple and a UNESCO World Heritage Site.
10	Lankatilaka Temple	Religious & Heritage	14 km	An ancient temple famous for its unique architecture and historical importance.
11	Gadaladeniya Temple	Religious & Heritage	13 km	A historic temple reflecting South Indian architectural influence.
12	Ambuluwawa Biodiversity Complex & Tower	Eco Tourism	24 km	A biodiversity reserve featuring an iconic observation tower and scenic landscapes.
13	National Museum of Kandy	Museum	0.5 km	A museum displaying Kandyan cultural artifacts and

				historical collections.
14	Commonwealth War Cemetery	Heritage	5 km	A well-maintained cemetery commemorating Commonwealth soldiers of World War II.
15	Kandy Viewpoint (Arthur's Seat)	Viewpoint	2 km	A popular viewpoint overlooking Kandy city, Kandy Lake, and the surrounding hills.

Table 2: Preliminary List of Places

1.9 Technology Stack

The proposed system will be developed using the following technology stack, selected for its industry relevance, strong ecosystem support, and suitability for building secure, scalable, and maintainable web applications:

- Frontend: Angular
- Backend Framework: ASP.NET Core Web API
- Programming Language: C#
- Database: SQL Server
- Development Tool: Visual Studio Code
- Version Control: Git and GitHub

Data access between the ASP.NET Core Web API and the SQL Server database will be implemented using Entity Framework Core, the standard object-relational mapping (ORM) framework for .NET applications. The Angular frontend will consume the backend through RESTful HTTP endpoints returning data in JSON format. Location-based features, including map display, distance calculation, and directions, will be provided through integration with Google Maps API. This technology stack provides a robust, well-documented, and cost-effective foundation for developing a responsive web application capable of managing tourist information and supporting one-day visit planning.

1.10 Functional Requirements

The functional requirements below describe the core services that the proposed system shall provide, grouped by user module.

ID	Module	Requirement	Description
FR1	Tourist	View Tourist Attractions	The system shall display a list of tourist attractions within the selected area.
FR2	Tourist	Search Tourist Attractions	Users shall be able to search for attractions using the place name.
FR3	Tourist	Filter by Category	Users shall be able to filter attractions by categories such as Religious & Heritage, Nature, Museum, Adventure, Recreation, Eco Tourism, and Viewpoint.
FR4	Tourist	View Attraction Details	The system shall display detailed information including the attraction name, description, category, opening hours, travel tips, images, approximate distance, and map location.
FR5	Tourist	View Location on Map	Users shall be able to view the location of each attraction using an interactive map.
FR6	Tourist	Creating a One-Day Visit Plan	Users shall be able to select multiple attractions and generate a personalized one-day travel itinerary.
FR7	Administrator	Administrator Login	The administrator shall log into the system

			using authorized credentials.
FR8	Administrator	Add Tourist Attractions	The administrator shall be able to add new tourist attraction information.
FR9	Administrator	Update Tourist Attractions	The administrator shall be able to edit existing tourist attraction details.
FR10	Administrator	Delete Tourist Attractions	The administrator shall be able to remove outdated or incorrect tourist attraction records.

1.11 Non-Functional Requirements

The non-functional requirements below describe the quality attributes that the proposed system shall satisfy.

ID	Category	Requirement Description
NFR1	Performance	The system should display search results and attraction details within a few seconds under normal operating conditions.
NFR2	Usability	The user interface should be simple, intuitive, and easy to navigate for both local and international tourists.
NFR3	Reliability	The application should provide accurate and consistent tourist information with minimal system errors.
NFR4	Security	Administrator's access shall be protected using secure authentication (e.g., ASP.NET Core Identity) to prevent unauthorized modifications.
NFR5	Availability	The web application should be accessible at any time through a standard web browser with an internet connection.
NFR6	Maintainability	The system should allow administrators to update tourist information without affecting existing functionality.
NFR7	Scalability	The system should support the addition of new tourist attractions and categories in future versions without requiring major structural changes.
NFR8	Compatibility	The application should function correctly on modern web browsers such as Google Chrome, Microsoft Edge, and Mozilla Firefox, and should

		provide responsive experience on desktop and mobile devices.
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1.12 Proposed Methodology

This project will be delivered using a structured, phase-based System Development Life Cycle (SDLC) combined with an iterative and incremental approach to implementation, allowing requirements to be validated and refined progressively rather than fixed entirely at the outset. This reflects standard business analysis practice of eliciting, analyzing, documenting, and validating requirements before and during solution design.

Requirements were elicited primarily through structured stakeholder consultation, conducted as a moderated online discussion with four group members acting as representative stakeholders (see Section 1.6 and Appendix A), supplemented by review of secondary sources on tourist attractions in the Kandy area [1]-[3]. The delivery approach consists of the following phases:

1. Requirement Elicitation and Analysis: Consultation with stakeholders to gather and analyses functional and non-functional requirements.
2. System Design: Design of the database schema, user interface wireframes, and the ASP.NET Core Web API contract.
3. Incremental Implementation: Development of the attraction catalogue and search/filter functions first, followed by map integration, the one-day itinerary planner, and finally the administrator module.
4. Testing and Validation: Verification of the implemented system against the functional and non-functional requirements defined in Sections 1.10 and 1.11.
5. Documentation and Deployment: Preparation of supporting documentation and deployment of the completed system for submission.

These SDLC phases are carried out within the module's weekly schedule shown in Section 1.13 and Appendix B, with solution and database design undertaken in Weeks 7-8 and implementation carried out in Weeks 9-11.

1.13 Risk Analysis

The table below identifies key risks to the successful delivery of the project, together with their likelihood, potential impact, and proposed mitigation strategies.

ID	Risk	Likelihood	Impact	Mitigation Strategy
R1	Incomplete or inaccurate tourist attraction data	Medium	Medium	Cross-verify details against official sources such as the Sri Lanka Tourism Development Authority, the Department of Wildlife Conservation, and the Department of Archaeology before publishing.
R2	Scope creep beyond the defined project scope	Medium	High	Adhere strictly to the scope and limitations defined in Section 1.4; proposed changes to be reviewed by the group before implementation.
R3	Limited prior experience with Angular, ASP.NET Core, and C#	Medium	Medium	Allocate early project time to prototyping and consulting official documentation before full-scale development.
R4	Delays due to uneven task distribution	Medium	Medium	Maintain a clear activity list (Appendix B) with defined

	among group members			responsibilities and regular progress reviews.
R5	Third-party API limitations (e.g., Google Maps API usage quotas)	Low	Medium	Use a development/test API key and monitor usage during testing.
R6	Unauthorized access to the administrator module	Low	High	Implement secure authentication and authorization using ASP.NET Core Identity.
R7	Fixed academic submission deadline	Medium	High	Follow the phased timeline in Section 1.13 and Appendix B, with buffer time allocated before final submission.

1.14 Expected Outcomes and Benefits

The successful completion of this project is expected to deliver the following outcomes and benefits:

- A fully functional, centralized web-based platform that allows tourists to search, filter, and view verified information about tourist attractions within a 25 km radius of Kandy.
- A reduction in the time and effort tourists currently spend gathering scattered information from multiple websites and social media platforms.
- A personalized, easy-to-use one-day itinerary planning feature that supports more efficient travel decisions.
- A secure administrator module that allows tourist information to be kept accurate and up to date over time.
- Practical, hands-on experience for the project team in requirement elicitation, stakeholder consultation, and full-stack web application development using Angular, ASP.NET Core Web API, C#, and SQL Server.
- A contribution, at a local level, to promoting tourism in the Kandy area by improving access to organized and reliable visitors' information.

1.15 Assumptions and Constraints

The following assumptions and constraints have been identified for this project:

Assumptions:

- Tourist attraction data (descriptions, opening hours, distances) will be sourced from publicly available and official information and verified before publication.
- Users will have access to a modern web browser and an active internet connection when using the system.
- Group members will remain available for stakeholder consultation and testing throughout the project timeline.

Constraints:

- The project must be completed within the fixed academic timeline defined in Section 1.13 and Appendix B.
- Development is limited to the technologies specified in Section 1.9 (Angular, ASP.NET Core Web API, C#, SQL Server).
- The system will only cover tourist attractions within approximately 25 km of Kandy, as defined in Section 1.4.

Conclusion

The proposed Local Tourist Day-Visit Planner and Information System addresses a clearly defined business problem: the absence of a centralized, reliable source of tourist information for one-day visit planning in the Kandy area. The system will allow tourists to search attractions, view detailed information, explore locations using an interactive map, and create personalized one-day travel itineraries, while a secure administrator module ensures that tourist information remains accurate, complete, and up to date. By combining a structured requirement elicitation process, a modern three-tier technology stack comprising Angular, ASP.NET Core Web API, C#, and SQL Server, and a phased, iterative delivery methodology, the project is expected to produce a functional, secure, and user-friendly system within the available project timeline, contributing to more efficient travel planning while promoting local tourism. Beyond its immediate deliverable, the project provides the team with practical experience in requirements analysis, stakeholder consultation, web application development, database management, and software engineering practice.

References

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Appendices

Appendix A: Stakeholder Consultation Evidence

Stakeholder consultation was carried out through a structured online discussion forum (ITE2953 - Programming Group Project, Group 39), comprising six discussion threads. These cover the problems tourists typically face (Discussion 1), the features expected of a tourism application (Discussion 2), the specific features required of a local day-visit planner (Discussion 3), features that should fall outside the project scope (Discussion 4), a final summary of the identified requirements (Discussion 5), and the initial requirement elicitation questionnaire that opened the consultation (Discussion 6).

Summary of Key Points - Expected Features of a Tourism Application

One of the discussions specifically addressed the question: 'What features or functionalities would you expect from a tourism application to make travelling easier, safer, and more enjoyable?' A summary of the responses received from the four group members consulted as stakeholders is provided below.

- James JBK (E2410439): Recommended that each attraction listing includes its name, a brief description, photographs, opening hours, entrance fees, and contact details.
- Werasekara SNDRWMMGJCB (E2410436): Suggested that every listing provide core identity (name, category), quick context (short description), and practical planning information (hours, entry fee, visit duration, distance/travel time, accessibility), along with photos, ratings, best time to visit, nearby transport, and filtering options.
- Seema AFZ (E2410442): Proposed displaying each attraction's exact location using Google Maps, together with the distance and estimated travel time from the user's location.
- Madhushan S (E2410438): Suggested including information about nearby facilities such as restaurants, cafes, parking, fuel stations, washrooms, and emergency services, and recommending other nearby attractions for a more enjoyable one-day trip.

These responses, together with the outcomes of the other discussion threads, were analyzed and used to shape the functional requirements and key features of the proposed system described in Sections 1.10 and 1.7. The complete discussion threads are reproduced below as supporting evidence.

Discussion 1: Problems Faced by Tourists When Visiting a New Area

ITE2953 - Programming Group Project 25S2

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SH_Grp_39

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Local Tourist Day-Visit Planner

Stakeholder Consultation – Requirement Elicitation for the Local Tourist Day-Visit Planner and Information System

Stakeholder Consultation – Discussion 2: Expected Features of a Tourism Application Message: ▶

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**Local Tourist Day-Visit Planner**
by [E2410439 James J.B.K.](#) - Wednesday, 5 August 2026, 8:46 PM

As informed by our project instructions, let's use this forum for our official stakeholder consultation.

Since we are all developing the same project individually, I thought it would be a good idea for each of us to lead one discussion topic. This way, everyone can actively participate, share their ideas, and help identify useful requirements for our individual projects.

Let's begin with our first discussion.

Question 1:

What problems do tourists usually face when visiting a new area for the first time?

Permalink Reply

**Re: Local Tourist Day-Visit Planner**
by [E2410433 Kishalini T.](#) - Wednesday, 5 August 2026, 10:02 PM

In my opinion, tourists usually face several challenges when visiting a new area for the first time. They may have difficulty finding reliable information about nearby attractions, opening hours, entrance fees, or the best time to visit. They may also struggle with navigation, planning an efficient one-day trip, and finding facilities such as parking, restaurants, or restrooms. Having all this information in one application would make their visit easier and more enjoyable.

Permalink Show parent Reply Export to portfolio

**Re: Local Tourist Day-Visit Planner**
by [E2410442 Seema A.F.Z.](#) - Wednesday, 5 August 2026, 10:06 PM

Hello James!

That is a nice question.

I think tourists often face several challenges when visiting a new place for the first time. Some common problems include difficulty finding reliable information about attractions, transportation, and accommodation. They may also struggle with language barriers, unfamiliar routes, safety concerns, and identifying trustworthy restaurants or local services. In addition, many tourists find it difficult to plan their itinerary efficiently, which can lead to wasted time and missed attractions. A system that provides accurate, up to date information and personalized recommendations could help solve many of these issues.

Permalink Show parent Reply

**Re: Local Tourist Day-Visit Planner**
by [E2410436 Werasekara S.N.D.R.W.M.M.G.J.C.B.](#) - Wednesday, 5 August 2026, 10:23 PM

These are the core frustrations independent travelers face: information overload without filtering, poor time estimation, unreliable data (hours, closures, reviews), cultural/language gaps, hidden costs, safety/transit uncertainty, offline access needs, choice paralysis among attractions, and missing insider knowledge that only locals typically have

Permalink Show parent Reply

**Re: Local Tourist Day-Visit Planner**
by [E2410438 Madhusan S.](#) - Wednesday, 5 August 2026, 10:36 PM

I agree with all of your points. In addition to those, I think tourists also find it difficult to locate nearby facilities such as restaurants, washrooms, parking areas, and fuel stations while travelling. Having all of this information available in one place would make it much easier for visitors to plan a comfortable and enjoyable one-day trip.

Permalink Show parent Reply

**Local Tourist Day-Visit Planner**
by [E2410439 James J.B.K.](#) - Wednesday, 5 August 2026, 11:12 PM

Thank you all for sharing your valuable ideas and insights. Based on your responses, it is clear that tourists need a single platform that provides reliable and well-organized information, along with essential travel details to help them plan their visits more effectively.

Your feedback has been very helpful in identifying the key requirements for our projects.

Now, let's move on to the next discussion topic.

Permalink Show parent Reply

Stakeholder Consultation – Requirement Elicitation for the Local Tourist Day-Visit Planner and Information System

Stakeholder Consultation – Discussion 2: Expected Features of a Tourism Application Message: ▶

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Figure 2: Stakeholder Consultation - Discussion 1

Discussion 2: Expected Features of a Tourism Application

ITE2953 - Programming Group Project 25S2

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ITE2953-25S2 > Stake Holder Analysis > SH_Grp_39 > Stakeholder Consultation – Discussion 2: Expected Features of a Tourism Application Message:

SH_Grp_39

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Stakeholder Consultation – Discussion 2: Expected Features of a Tourism Application Message:

Local Tourist Day-Visit Planner Stakeholder Consultation – Discussion 03

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**Stakeholder Consultation – Discussion 2: Expected Features of a Tourism Application Message:**
by [E2410433 Kishalani T](#) · Thursday, 6 August 2026, 8:02 PM

As part of our stakeholder consultation, let's continue with our second discussion topic.

In our previous discussion, we talked about the common problems tourists face when visiting a new area for the first time. Now, let's focus on the features and functionalities that could help solve those challenges.

Question 2:

What features or functionalities would you expect from a tourism application to make travelling easier, safer, and more enjoyable?

[Permalink](#) [Reply](#) [Export to portfolio](#)

**Re: Stakeholder Consultation – Discussion 2**
by [E2410439 James JBK](#) · Thursday, 6 August 2026, 8:33 PM

I think each tourist attraction should provide essential information such as its name, a brief description, photographs, opening and closing hours, entrance fees (if applicable), and contact details. This would help visitors understand what to expect before planning their visit.

[Permalink](#) [Show parent](#) [Reply](#)

**Re: Stakeholder Consultation – Discussion 2: Expected Features of a Tourism Application Message:**
by [E2410442 Seema AFZ](#) · Thursday, 6 August 2026, 8:54 PM

Hi Kishalani

Thank you for the question.

In my opinion, The system should also display the exact location using Google Maps, together with the distance from the user's location and estimated travel time.

[Permalink](#) [Show parent](#) [Reply](#)

**Re: Stakeholder Consultation – Discussion 2: Expected Features of a Tourism Application Message:**
by [E2410436 Werasekara SDRWMMGJCB](#) · Thursday, 6 August 2026, 9:06 PM

For every traveler's needs Every attraction listing needs core identity (name, category), quick context (short description), and practical planning info: hours, entry fee, visit duration, distance/travel time, and accessibility. Add 1–2 photos and a rating for quick trust-building, plus best time to visit and nearby transport for smoother planning.

It would be much easier for tourists if you included details like this, wouldn't it ?

It would be great if you could also give them a chance to filter this as well ?

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**Re: Stakeholder Consultation – Discussion 2: Expected Features of a Tourism Application Message:**
by [E2410438 Madhushan S](#) · Thursday, 6 August 2026, 9:37 PM

In addition to the basic attraction details, I think the system should display information about nearby facilities such as restaurants, cafes, parking areas, fuel stations, washrooms, and emergency services. It would also be helpful to suggest other nearby attractions so that tourists can plan a more enjoyable and efficient one-day trip.

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**Re: Stakeholder Consultation – Discussion 2: Expected Features of a Tourism Application Message:**
by [E2410433 Kishalani T](#) · Thursday, 6 August 2026, 9:51 PM

Thank you everyone for your valuable ideas and active participation in this second stakeholder consultation.

From your feedback, the key features expected in a tourism application include:

- * Detailed attraction information (name, description, photos, opening hours, entrance fees, and contact details).
- * Google Maps integration with location, distance, and estimated travel time.
- * Ratings, accessibility details, best time to visit, nearby transport, and filtering options.
- * Information about nearby facilities such as restaurants, cafes, parking, fuel stations, washrooms, emergency services, and recommendations for nearby attractions.

Your suggestions have provided valuable insights into the features that would make a tourism application more useful, convenient, and enjoyable for travelers. These requirements will be considered during the system design and development process.

Thank you once again for your time and contributions. We appreciate your participation and look forward to your involvement in our next stakeholder consultation discussion.

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Local Tourist Day-Visit Planner Stakeholder Consultation – Discussion 03

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Figure 3: Stakeholder Consultation - Discussion 2

Discussion 3: Key Features for a Local Tourist Day-Visit Planner

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Stakeholder Consultation – Discussion 03

[◀ Stakeholder Consultation – Discussion 2: Expected Features of a Tourism Application Message:](#)[Stakeholder Consultation – Discussion 04 ▶](#)

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**Stakeholder Consultation – Discussion 03**
by [E2410442 Seema AFZ](#) - Thursday, 6 August 2026, 9:56 PM

Hello everyone !

As part of our stakeholder consultation I think it is the time to move on to our third discussion topic.

In our previous discussion we identified the features that could improve the travel experience. Now, let's discuss the essential features that should be included in a Local Tourist Day-Visit Planner.

Question 03:

What features would you expect from a Local Tourist Day-Visit Planner to help tourists organize and manage their one-day trip efficiently?

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**Re: Stakeholder Consultation – Discussion 03**
by [E2410439 James JBK](#) - Thursday, 6 August 2026, 10:09 PM

I think the system should include a search function, category filters, and location-based recommendations so users can quickly find attractions that match their interests. It would also be helpful if users could view key information and plan their day more efficiently from a single interface.

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**Re: Stakeholder Consultation – Discussion 03**
by [E2410433 Kishalani T](#) - Thursday, 6 August 2026, 10:30 PM

In my opinion, a Local Tourist Day-Visit Planner should help tourists plan their day with minimum effort. It should include personalized itinerary suggestions based on available time and interests, an interactive map with navigation, details about attractions (opening hours, entrance fees, and ratings), nearby restaurants and rest areas, estimated travel time between locations, and a budget tracker. It would also be useful to receive weather updates, safety information, and notifications about traffic or attraction closures. These features would help tourists make the most of their one-day trip in a convenient and organized way.

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**Re: Stakeholder Consultation – Discussion 03**
by [E2410436 Werasekara SMDRWMMGJCB](#) - Thursday, 6 August 2026, 10:44 PM

There are a few key features to look for in a local travel day-trip planner. First, a day-trip builder automatically builds a logical route based on selected locations, opening hours, and travel time. Filter/search features should be available by category, budget, distance, kid-friendly A map view should show the route between stops.

In addition, the itinerary should be able to be saved/shared, the map/plan should be accessible in offline mode, and the rest of the plan should be reshuffled in real-time if a stop is delayed. A budget calculator that estimates the total cost of the day (transport + tickets + food) and a genuine local tips section rather than crowd reviews are also important.

I think these things should be in a Visit Planner

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**Re: Stakeholder Consultation – Discussion 03**
by [E2410438 Madhusan S](#) - Thursday, 6 August 2026, 11:20 PM

I think the system should have an administrator panel where authorized users can add new tourist attractions, update existing information, upload photos, and remove outdated content. This will ensure that the information provided to tourists remains accurate, reliable, and up to date.

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**Re: Stakeholder Consultation – Discussion 03**
by [E2410442 Seema AFZ](#) - Thursday, 6 August 2026, 11:26 PM

Thank you everyone for your valuable suggestions and active participation. These features would definitely improve the overall user experience.

Based on this discussion, the key features expected from a Local Tourist Day-Visit Planner include a search function, category filtering, one-day itinerary planning, Google Maps integration, route guidance, and administrator management of tourist attraction information.

These suggestions provide a clear understanding of the functional requirements that should be considered when developing the system. I sincerely appreciate everyone's valuable contributions and participation in this discussion.

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[◀ Stakeholder Consultation – Discussion 2: Expected Features of a Tourism Application Message:](#)[Stakeholder Consultation – Discussion 04 ▶](#)

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Figure 4: Stakeholder Consultation - Discussion 3

Discussion 4: Features Outside the Project Scope

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[◀ Stakeholder Consultation – Discussion 03](#)[Stakeholder Consultation 05 - Feedback summary & Identified Requirements ▶](#)

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**Stakeholder Consultation – Discussion 04**
by [E2410436 Werasekara SDRWMMGJCB](#) · Friday, 7 August 2026, 9:35 PM

Hello everyone !

In our previous discussion, we talked about helping tourists to efficiently complete their day trip.

The fourth topic is to study the features that should not be included when considering the scope of the project. Let's start the discussion now.

Question 04

When considering the scope of the project, what features should not be included ?

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**Re: Stakeholder Consultation – Discussion 04**
by [E2410433 Krishalani T](#) · Friday, 7 August 2026, 9:45 PM

In my opinion, the application should remain focused on planning and managing one-day local trips. Therefore, features such as hotel booking, flight booking, online shopping, social media messaging, ride-sharing services, and multi-day tour package management should not be included. These features are outside the scope of the project and would make the application more complex than necessary.

Instead, the application should focus on its core purpose by providing day-trip planning, attraction information, navigation, itinerary management, budget estimation, and location-based recommendations. Keeping the scope limited to these essential features will make the application simpler, more user-friendly, and better aligned with the project objectives.

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**Re: Stakeholder Consultation – Discussion 04**
by [E2410439 James JBK](#) · Friday, 7 August 2026, 9:50 PM

Since the project is designed as a Local Tourist Day-Visit Planner and Information System, features such as hotel booking and multi-day trip planning are beyond its scope. The main focus should remain on helping tourists plan a simple and efficient one-day visit by providing accurate information and travel guidance.

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**Re: Stakeholder Consultation – Discussion 04**
by [E2410442 Seema AFZ](#) · Friday, 7 August 2026, 10:00 PM

Hello !

I believe transport reservation features should not be included in this project. The main objective of the system is to help tourists plan a one-day visit by providing useful information about tourist attractions. Including transport reservation or booking services would make the system more complex than required and go beyond the intended project scope. Therefore, the focus should remain on providing accurate information and effective trip planning.

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**Stakeholder Consultation – Discussion 04**
by [E2410438 Madhusan S](#) · Friday, 7 August 2026, 10:10 PM

I agree with the points discussed. Since this project is intended to help tourists plan a one-day visit, it should focus on providing accurate attraction details and practical planning tools. Adding features like hotel bookings, online payments, or transport reservations would increase the complexity of the system and shift its focus away from its main objective.

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**Re: Stakeholder Consultation – Discussion 04**
by [E2410436 Werasekara SDRWMMGJCB](#) · Friday, 7 August 2026, 10:29 PM

I don't think these should be included

focuses solely on single-day visit planning, so several adjacent features are intentionally out of scope. Hotel/flight booking belongs to OTAs, not a day-trip planner. Multi-day itineraries would require persistence and accommodation logic beyond our core use case. Social features (feeds, following, comments) and full review systems add scope creep we'll curate existing ratings instead of building our own. . Real-time crowd tracking needs data partnerships or hardware we don't have. Without open reviews/photos, content moderation isn't needed either.

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[◀ Stakeholder Consultation – Discussion 03](#)[Stakeholder Consultation 05 - Feedback summary & Identified Requirements ▶](#)

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Figure 5: Stakeholder Consultation - Discussion 4

Discussion 5: Feedback Summary and Identified Requirements

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Stakeholder Consultation 05 - Feedback summary & Identified Requirements

Stakeholder Consultation - Discussion 04

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Stakeholder Consultation 05 - Feedback summary & Identified Requirements
by [E2410438 Madhushan S](#) - Friday, 7 August 2026, 11:17 PM

Thank you, everyone, for your active participation and valuable insights throughout this stakeholder consultation.

Based on our discussions, we have identified the following key functional requirements for the **Local Tourist Day-Visit Planner and Information System**:

- A centralized platform with comprehensive tourist attraction information
- Search functionality and category-based filtering
- Google Maps integration with directions and estimated travel time
- Attraction descriptions, photographs, opening hours, and entrance fees
- Travel tips and visitor guidelines
- Information about nearby restaurants, parking areas, washrooms, fuel stations, and other facilities
- One-day itinerary planning to help tourists organize their visits efficiently
- An administrator panel to manage and update attraction information

We also agreed that the following features are outside the scope of the current project:

- Hotel and accommodation booking
- Online payment facilities
- Transport reservation services

Overall, the discussion helped us identify the essential requirements and define a clear project scope. These stakeholder suggestions will be valuable in developing a user-friendly and practical system for tourists planning one-day visits.

Thank you once again for your valuable contributions. I appreciate everyone's participation and look forward to seeing our individual projects develop based on these ideas.

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Stakeholder Consultation - Discussion 04

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Figure 6: Stakeholder Consultation - Discussion 5

Discussion 6: Requirement Elicitation Questionnaire (Initial Consultation Invitation)

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Stakeholder Consultation – Requirement Elicitation for the Local Tourist Day-Visit Planner and Information System

Local Tourist Day-Visit Planner

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Stakeholder Consultation – Requirement Elicitation for the Local Tourist Day-Visit Planner and Information System

by [E2410433 Kirshalani T](#) · Monday, 3 August 2026, 11:35 PM

Dear Group Members,

I hope you are all doing well.

As part of my **ITE2953 Programming Group Project**, I am conducting a stakeholder consultation to understand the needs and expectations of potential users before finalizing the system requirements for my **Local Tourist Day-Visit Planner and Information System**.

The purpose of this application is to help visitors discover and plan a one-day trip to places of interest within a **25 km radius**. Before I prepare my **Software Requirements Specification (SR3)**, I would like to gather your valuable feedback.

1. Understanding the Problem

From your experience, what difficulties do tourists usually face when visiting a new area?

For example:

- Finding nearby attractions
- Lack of reliable information
- Difficulty planning a one-day trip
- Not knowing opening hours or travel distance

Do you think there are any other common problems?

2. Information Required

What information do you think should be available for each tourist place?

Some examples are:

- Place Name
- Description
- Category
- Opening & Closing Hours
- Distance
- Google Maps Location
- Travel Tips
- Entry Fee
- Contact Details
- Photos

Is there any other information that would be useful?

3. Categories of Places

Which categories should be included in the system?

For example:

- Historical Places
- Religious Places
- Parks & Gardens
- Nature Attractions
- Beaches
- Museums
- Restaurants & Cafes
- Cultural Places

Would you recommend any additional categories?

4. Visitor Features

What features would make this application more useful for visitors?

Examples:

- Search places
- Filter by category
- View place details
- Google Maps
- Route guidance
- One-day visit planner
- Favourite places

Which features do you think are most important?

5. Future Improvements

The current project focuses on **tourist information** and **visit planning**. Features such as **hotel booking**, **transport booking**, and **online payment** are outside the current project scope.

If the system is improved in the future, what additional features would you like to see?

6. Final Feedback

Please share any other ideas or suggestions that could improve this application.

Your feedback will help me identify user needs, validate the project requirements, and prepare a better **Software Requirements Specification (SR3)**.

Thank you for your valuable time and support. I look forward to your suggestions.

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Local Tourist Day-Visit Planner

Figure 7: Stakeholder Consultation - Discussion 6

Appendix B: Project Plan

The Gantt chart below presents the project plan aligned with the ITE2953 module's 14-week teaching schedule, from course introduction and problem specification through to final report evaluation. Development activities described in Section 1.12 (solution design, database design, and implementation) are carried out primarily during Weeks 7-11.

No.	Activity	Intro	1	2	3	4	5	6	7	8	9	10	11	12	Mid-Exam	13	14
1	Course Introduction																
2	Introduction to Project Module																
3	Problem Specification																
4	How to Write a Project Proposal																
5	Requirement Analysis (SRS)																
6	Independent Work on Preparing SRS																
7	Peer Evaluation of SRS																
8	Solution Design with Diagrammatic Techniques using CASE Tools																
9	Database Design with ER Diagram																
10	Implementation using Version Control Systems																
11	Project Implementation																
12	Project Implementation and Report Preparation																
13	Testing Results and Evaluation of the Project																
14	Mid-Semester Exam																

15	Testing Results and Evaluation of the Project																
16	Report Evaluation																